

A Drag Coefficient Estimation

The drag coefficients relative to the rocket's Mach number throughout flight, both with and without the motor producing thrust, are required for accurate simulations.

These can be measured experimentally or can be determined using computational methods. This section explains how to determine them experimentally using RAS Aero II (*Welcome to RASAero.Com* 2026), a combined aerodynamic and flight simulation program that can accurately predict subsonic-to-hypersonic aerodynamic coefficients for rockets and has been calibrated against NACA and NASA wind tunnel model data. RAS Aero II is also widely used in the amateur and student rocketry community and can be used to extract aerodynamic coefficients for use in other tools.

Step 1: OpenRocket Simulation

Create an accurate representation of the rocket's geometry in OpenRocket. This can be done directly in RAS Aero II, skipping steps 1 and 2, but it is generally easier to do in OpenRocket.

Step 2: Export to RAS Aero II

Export the OpenRocket design to RASAeroII from within OpenRocket ("File" > "Export As" > "RAS Aero II") as seen in fig. A.1. If prompted, you may need to remove or amend some unsupported features from OpenRocket before this is successful. This won't affect the drag coefficients unless you change external geometry.

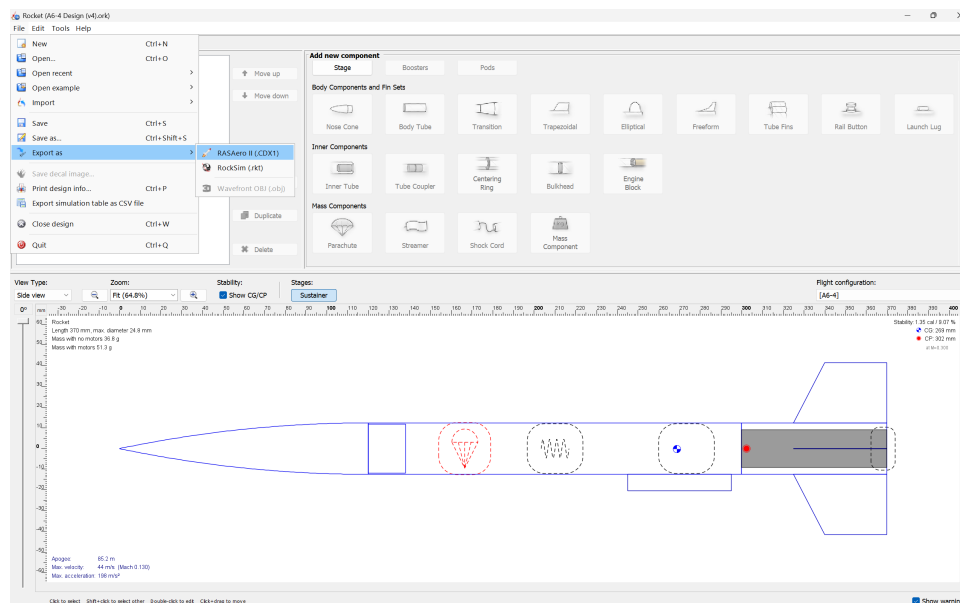


Figure A.1: Exporting to RAS Aero II from OpenRocket

Step 3: Open exported RAS Aero II file

Open RAS Aero II and then open the exported RAS Aero II file ("File" > "Open"). Once opened, confirm that the dimensions and geometry match those of the OpenRocket model.

Step 4: Add motor

Check that the correct motor has been selected (no motor may be selected) as shown in the blue circle of fig. A.2.

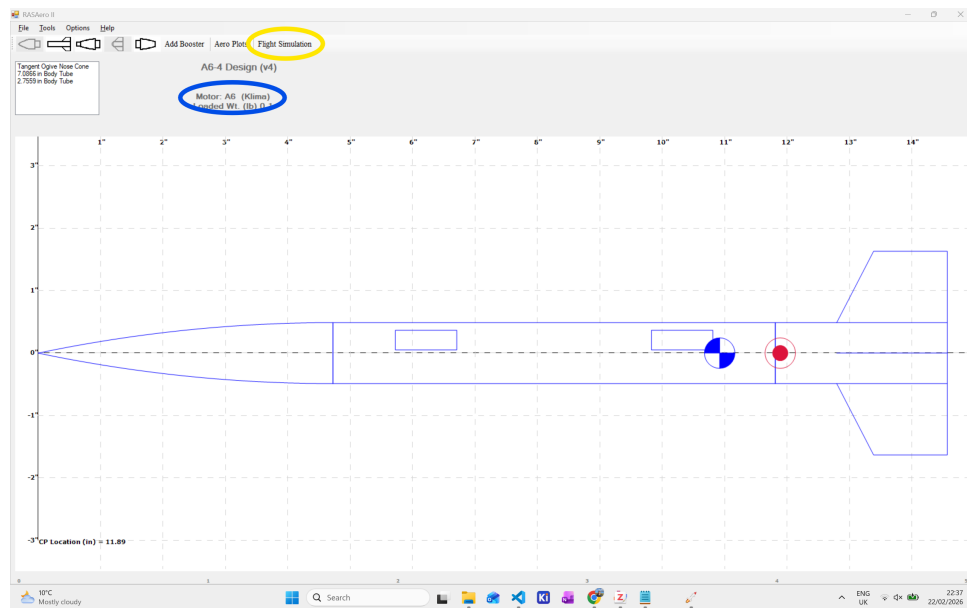


Figure A.2: Checking motor selection in RAS Aero II. Blue: Checking motor selection. Yellow: Opening flight simulations window

If the incorrect or no motor is selected, you should add the correct motor's thrust curve (.eng RASP file), which is available from thrustcurve.org (*Rocket Motor Data • ThrustCurve 2026*) for most commercial motors, into RAS Aero II by selecting "File" > "Select Motor File". You then need to load it into the rocket by going to the "Flight Simulation" tab as seen in the yellow circle of fig. A.2 and then add this motor by selecting the motor slot and then the motor from the dropdown as shown in fig. A.3 in blue and yellow circles respectively. The flight simulation window can then be closed.

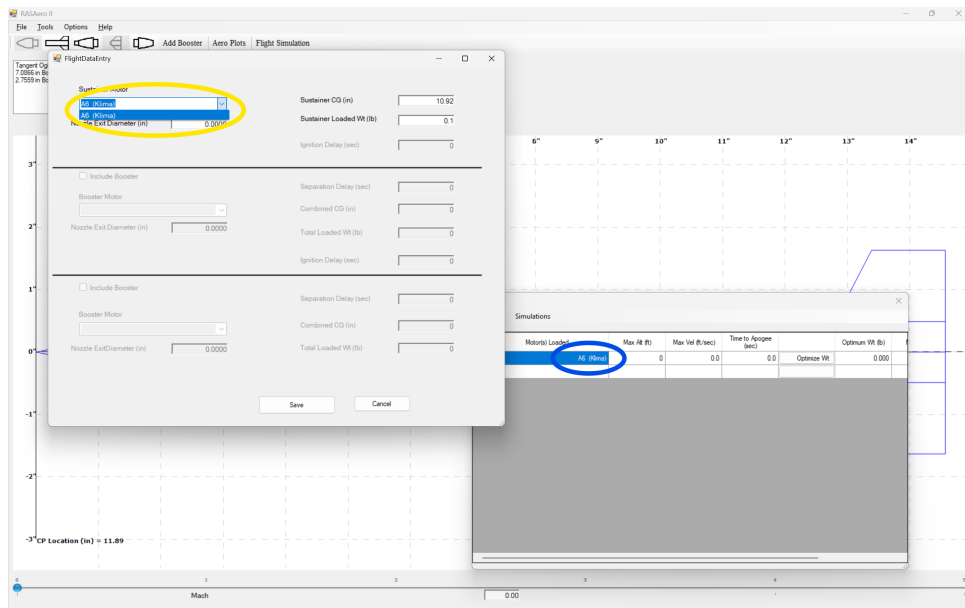


Figure A.3: Selecting a motor to use in RAS Aero II. Blue: Select motor slot. Yellow: Select which motor to use from the dropdown

Step 5: Create Aero Plots

Open the aerodynamic plots window by selecting the "Aero Plots" button seen in a blue circle in fig. A.4 and then export this data to a CSV file as seen ("File" > "Export" > "To CSV File").

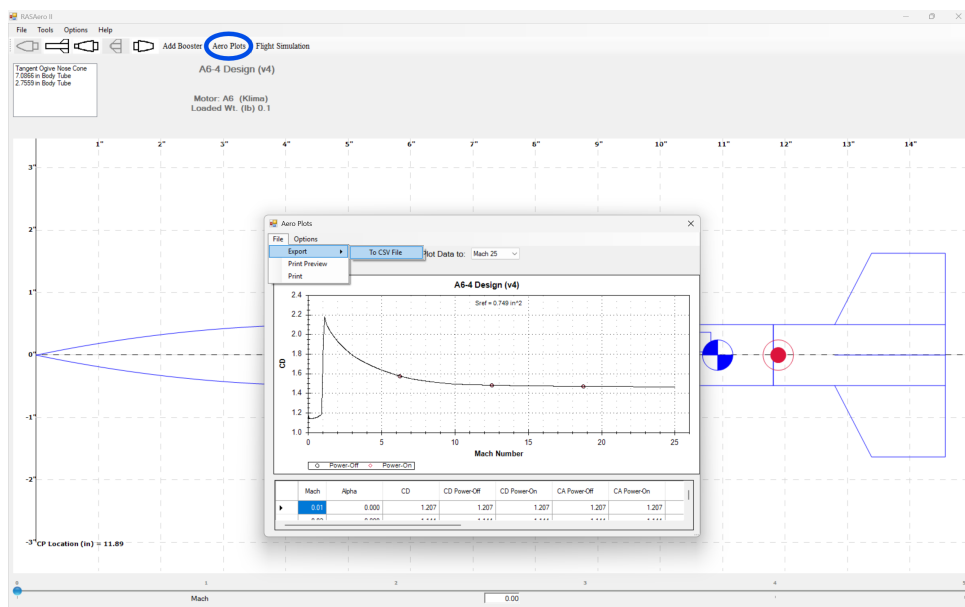


Figure A.4: Exporting aerodynamic plots from RAS Aero II. Blue: Opening the aero plots window.

Step 6: Extract useful data from csv

Use the `extract_drag_curves.py` script to extract 2 files (`powerOnDragCurve.csv` and `powerOffDragCurve.csv`) from the RAS Aero II exported file in the same directory.

```
$ python .\extract_drag_curves.py --input .\data\rockets\lpr\RASAeroOut.CSV
Extracted 'Mach' and 'CD Power-On' to C:\Users\rufus\OneDrive - The University of Manchester
\Documents\University\Year 3\Individual Project\code\models\data\rockets\lpr\
powerOnDragCurve.csv
Extracted 'Mach' and 'CD Power-Off' to C:\Users\rufus\OneDrive - The University of
Manchester\Documents\University\Year 3\Individual Project\code\models\data\rockets\lpr\
powerOffDragCurve.csv
```

B Inertia Approximation

The moments of inertia of a rocket are required for accurate simulations, as they heavily influence its stability and dynamics. These are usually measured experimentally or estimated from a detailed CAD model of the vehicle. However, these methods are often not possible or take too much time, such as during the early design of a rocket.

This section explains how to estimate the approximate moment or inertias of a rocket based upon its mass and geometry using OpenRocket. This simple method assumes the centre of inertia is equal in the pitch and yaw axes and ignores the moment of inertia of the motors, as this is calculated separately within the model. It also assumes that an OpenRocket model of your rocket already exists and that you are familiar with OpenRocket.

Step 1: Add empty motor to OpenRocket

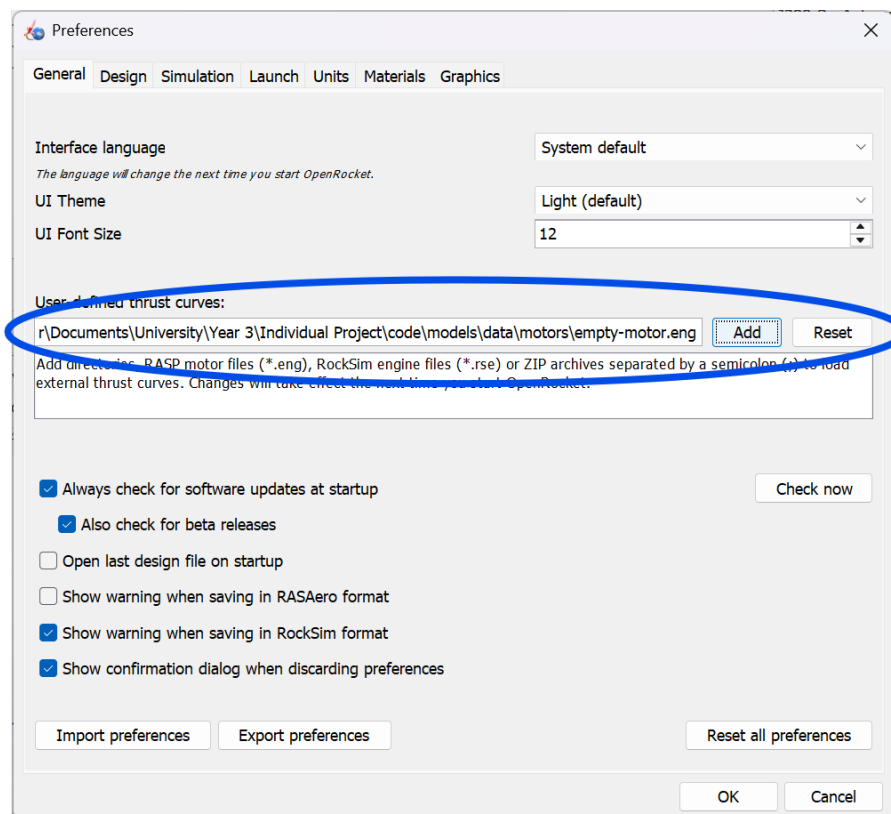


Figure B.1: Adding a motor to OpenRocket

Download the empty motor file from empty-motor.eng or create a mass-less (or near mass-less) motor yourself.

Open your OpenRocket model of the rocket. Navigate to user preferences ("Edit" > "Preferences") and then add the empty motors .eng file to OpenRocket as seen in fig. B.1 and then exit this menu.

Step 2: Add empty motor simulation

Create a new motor configuration with the new motor in the "Motors & Configuration" tab.

Step 3: Simulate and export

Simulate the flight with the new motor configuration and export the results to a CSV file with the "Plot / Export" button. Using the "Export Data" tab, make sure to include "Longitudinal moment of inertia" and "Rotational moment of inertia" as variables to export and select "Include flight events" as seen in fig. B.2. Press "Export" to save the data as a CSV file.

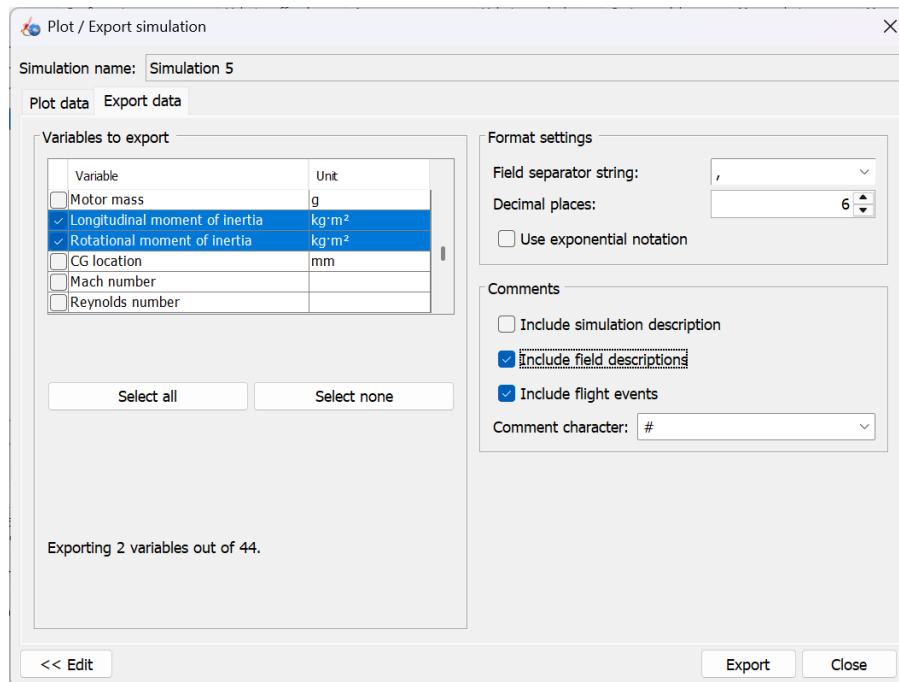


Figure B.2: Exporting flight data from OpenRocket

Step 4: Extract inertia values from simulations

Using the CSV files exported from OpenRocket, find the first row after motor burnout (or launch if there is no burnout event) and note the values of moment of inertia using the below relations about the dry centre of mass:

$$\text{Pitch Axis} = \text{Yaw Axis} = \text{Rotational Moment of Inertia} = I_{11} = I_{22}$$

$$\text{Roll Axis} = \text{Longitudinal Moment of Inertia} = I_{33}$$